Harshit Mangal

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Education

MBM University Jodhpur, Rajasthan
Bachelor of Technology in Information Technology — **CGPA: 7.2** Aug. 2023 – May 2027

Projects

ShopSphere — Live Demo — Frontend — Backend — *React, Node.js, Express, MongoDB, Socket.io, Redis, Redux Toolkit* 2026

- Built an AI-driven B2C e-commerce platform integrating a **Google Gemini 2.5 Flash** assistant (LangChain) for catalog queries and real-time translations, backed by a **resilient Redis cache with MongoDB failover** (≤2ms latency).
- Architected a real-time **Socket.io support chat** with room-based user-to-admin isolation and MongoDB aggregation pipelines for dynamic active thread management.
- Integrated dynamic multi-currency payments (**Razorpay/PayPal**) via IP Geolocation, social Group Buying with Razorpay HMAC verification, and secure JWT/OAuth sessions.

AI Document Assistant (Offline RAG) — GitHub Repo — *Python, Streamlit, Ollama (LLaMA 3.2), FAISS, SentenceTransformers, EasyOCR, PyMuPDF* 2025

- Architected a 100% offline, air-gapped RAG system using local **Ollama (LLaMA 3.2 3B)** and **SentenceTransformers**, eliminating external API costs and zero data leakage.
- Engineered a hybrid document parser (**PyMuPDF, python-docx**) with GPU-accelerated **EasyOCR** fallback for scanned PDFs and image-based documents.
- Implemented in-memory/disk **FAISS vector search** with 800-char semantic chunking and LLM query expansion, enabling sub-second retrieval and multi-user LAN hosting.

Wanderlust — Live Demo — Frontend — Backend — *React, Node.js, Express, MongoDB, Socket.io, Mapbox GL JS, Cloudinary* 2025

- Developed a full-stack geospatial rental booking SaaS using decoupled **React/Vite (SPA)** and **Node.js/Express REST APIs** with custom middleware route protection.
- Integrated **Mapbox GL JS & Geocoding API** for dynamic map plotting, localized nightly rate formatting (INR/USD), and real-time Socket.io guest-to-host messaging stored in MongoDB.
- Engineered **Cloudinary** media upload pipelines with dynamic compression and lazy loading, optimizing load speeds by 40% and image payload sizes by 60%.

Experience

Frontend Web Development Intern June 2025 – July 2025
CodTech IT Solutions Pvt. Ltd. *Remote*

- Independently built 3 production-grade projects: a real-time chat application, an interactive quiz app, and a responsive e-learning platform UI within a 2-month timeline.
- Implemented cross-browser compatible, mobile-first UI components using HTML, CSS, and JavaScript; integrated REST APIs and followed Git collaborative workflows.

Technical Skills

Languages: Python, C++, C, Java, JavaScript, SQL
Web & Frameworks: HTML, CSS, React, Redux Toolkit, Node.js, Express, Streamlit, Tailwind CSS, EJS, Mongoose, JWT, Socket.io
AI/ML & Vector DBs: RAG Architecture, Ollama (LLaMA 3.2), FAISS, SentenceTransformers, EasyOCR, PyMuPDF, Prompt Engineering
Databases & Cloud: SQLite, MySQL, MongoDB, Redis, Git, GitHub, Docker (Basics), Postman, Render, Vercel, Heroku, Linux
CS Fundamentals: Data Structures & Algorithms, OOPs, DBMS, Operating Systems (OS), RESTful APIs, MVC Architecture

Achievements

LeetCode Knight Badge: Earned prestigious **Knight Badge** with a Contest Rating of **1905** (Top 5% globally, 1350+ solved).
GeeksforGeeks College Rank 1: Secured **Rank 1** across MBM University on GeeksforGeeks with 1100+ problems solved.

Coding Profiles

LeetCode: Profile (1350+ Solved, Rating: 1905) **GeeksforGeeks:** Profile (1100+ Solved, College Rank 1)
NeetCode: Profile (Focused DSA Patterns) **InterviewBit:** Profile (Consistent Practice on DP & Graphs)

Certifications

160 Days of Problem Solving – GeeksforGeeks (2025)
HackerRank Certifications: Basic Problem Solving & Intermediate Problem Solving – HackerRank (2026)